# Backstepping

# Why?

- MRAC requires model matching conditions

$$\boxed{\begin{aligned} A + B\Lambda K_x^T &= A_m \\ B\Lambda K_r^T &= B_m \end{aligned}}$$

- Example that violates matching
  - System: $\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{A} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}_{b} u$
  - Reference model: $\begin{pmatrix} \dot{x}_1^m \\ \dot{x}_2^m \end{pmatrix} = \underbrace{\begin{pmatrix} -1 & 1 \\ 0 & -2 \end{pmatrix}}_{A_m} \begin{pmatrix} x_1^m \\ x_2^m \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} r$

Matching conditions don't hold

$$A - A_m = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \neq b\, k_x^T = \begin{pmatrix} 0 & 0 \\ * & * \end{pmatrix}$$

# Control Tracking Problem

- Consider 2nd order ***cascaded*** system

$$\boxed{\begin{aligned}\dot{x}_1 &= f_1(x_1) + g_1(x_1)\,x_2\\ \dot{x}_2 &= f_2(x_1, x_2) + g_2(x_1, x_2)\,u\end{aligned}}$$

- Control goal
  - Choose $u$ such that: $\boxed{x_1(t) \to x_1^{com}(t), \quad \text{as } t \to \infty}$
- Assumptions
  - All functions are known
  - $g_i \neq 0$ does not cross zero
- Example: AOA tracking

$$\boxed{\begin{cases}\dot{\alpha} = \underbrace{-L_\alpha(\alpha)\overbrace{\alpha}^{x_1}}_{f_1} + \underbrace{1}_{g_1}\overbrace{q}^{x_2}\\ \dot{q} = \underbrace{M_0(\alpha, q)}_{f_2} + \underbrace{1}_{g_2}\underbrace{\dot{q}_{cmd}}_{u}\end{cases}}$$

# Backstepping Design

- Introduce pseudo control: $\boxed{x_2^{com} = x_2^{com}(t)}$
- Rewrite the 1st equation:

$$\boxed{\dot{x}_1 = f_1(x_1) + g_1(x_1)\,x_2^{com} + g_1(x_1)\underbrace{\left(x_2 - x_2^{com}\right)}_{\Delta x_2}}$$

- Dynamic inversion using pseudo control:

$$\boxed{x_2^{com} = \frac{1}{g_1(x_1)}\left(\dot{x}_1^{com} - f_1(x_1) - k_1\,\Delta x_1\right)}$$

- 1st state error dynamics: $\boxed{\Delta\dot{x}_1 = -k_1\,\Delta x_1 + g_1(x_1)\,\Delta x_2}$

# Backstepping Design
(continued)

- Dynamic inversion using actual control

$$\boxed{u = \frac{1}{g_2\left(x_1, x_2\right)}\left(\dot{x}_2^{com} - f_2\left(x_1, x_2\right) - k_2\,\Delta x_2 - g_1\left(x_1\right)\Delta x_1\right)}$$

- 2nd state error dynamics

$$\boxed{\Delta\dot{x}_2 = -k_2\,\Delta x_2 - g_1\left(x_1\right)\Delta x_1}$$

- *<u>Asymptotically stable</u>* error dynamics

$$\boxed{\begin{pmatrix}\Delta\dot{x}_1\\ \Delta\dot{x}_2\end{pmatrix} = \begin{pmatrix}-k_1 & g_1\left(x_1\right)\\ -g_1\left(x_1\right) & -k_2\end{pmatrix}\begin{pmatrix}\Delta x_1\\ \Delta x_2\end{pmatrix}}$$

nonlinear system

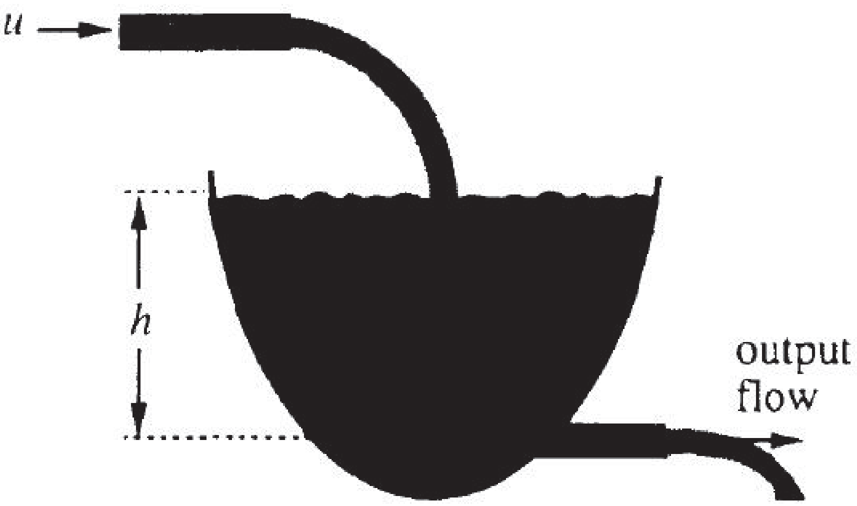
$V(\Delta x_1,\Delta x_2)=(\Delta x_1)^2+(\Delta x_2)^2$

- Conclusion: $\boxed{x_i\left(t\right) \to x_i^{com}\left(t\right), \quad \text{as } t \to \infty}$

# Feedback Linearization

# Feedback Linearization

- **The main idea is:** algebraically **transform** a nonlinear system dynamics into a (fully or partly) linear one, so that linear control techniques can be applied.
- In its simplest form, feedback linearization **cancels the nonlinearities** in a nonlinear system so that the closed-loop dynamics is in a linear form.
- **Example:** Controlling the fluid level in a tank
  - Objective: controlling of the level $h$ of fluid in a tank to a specified level $h_d$, using control input $u$
  - the initial level is $h_0$.




Fluid level control in a tank

# Example Cont'd

- The dynamics:

$$A(h)\dot{h}(t) = u - a\sqrt{2gh}$$

  where $A(h)$ is the cross section of the tank and $a$ is the cross section of the outlet pipe.
- Choose $u = a\sqrt{2gh} + A(h)v \rightsquigarrow \dot{h} = v$
- Choose the equivalent input $v$: $v = -\alpha\tilde{h}$ where $\tilde{h} = h(t) - h_d$ is error level, $\alpha$ a pos. const.
- $\therefore$ resulting closed-loop dynamics: $\dot{h} + \alpha\tilde{h} = 0 \Rightarrow \tilde{h} \to 0$ as $t \to \infty$
- The actual input flow: $u = a\sqrt{2gh} - A(h)\alpha\tilde{h}$
  - First term provides output flow $a\sqrt{2gh}$
  - Second term raises the fluid level according to the desired linear dynamics
- If $h_d$ is time-varying: $v = \dot{h}_d(t) - \alpha\tilde{h}$
  - $\therefore$ $\tilde{h} \to 0$ as $t \to \infty$

- Canceling the nonlinearities and imposing a desired linear dynamics, can be simply applied to a class of nonlinear systems, so-called companion form, or **controllability canonical form**:
- A system in companion form:

$$x^{(n)}(t) = f(\mathbf{x}) + b(\mathbf{x})u \tag{1}$$

  - $u$ is the scalar control input
  - $x$ is the scalar output;$\mathbf{x} = [x, \dot{x}, ..., x^{(n-1)}]$ is the state vector.
  - $f(x)$ and $b(x)$ are nonlinear functions of the states.
  - no derivative of input $u$ presents.
- (1) can be presented as controllability canonical form

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} x_2 \\ \vdots \\ x_n \\ f(x) + b(x)u \end{bmatrix}$$

- for nonzero $b$, define control input: $u = \frac{1}{b}[v - f]$

# Feedback Linearization

- $\therefore$ the control law:

$$v = -k_0 x - k_1 \dot{x} - \ldots - k_{n-1} x^{(n-1)}$$

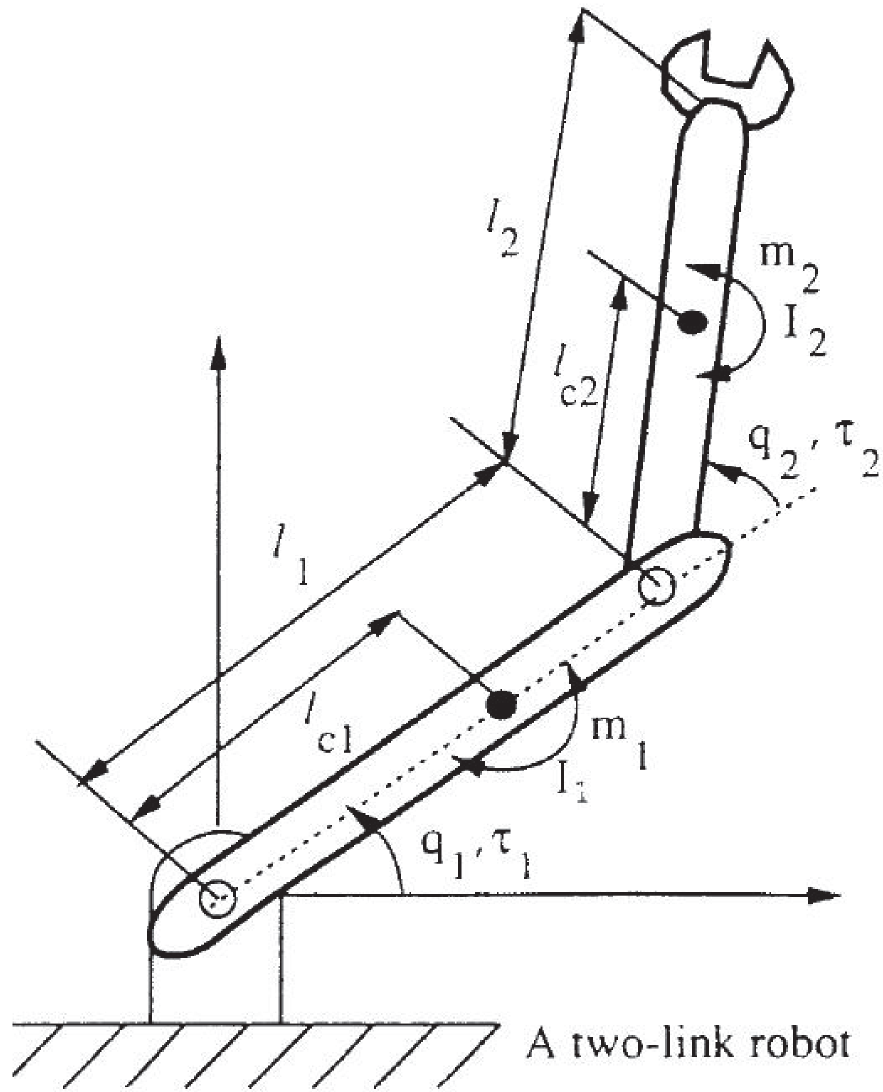
  - $k_i$ is chosen s.t. the roots of $s^n + k_{n-1}s^{n-1} + \ldots + k_0$ are strictly in LHP.
- **Thus**: $x^{(n)} + k_{n-1}x^{(n-1)} + \ldots + k_0 = 0$ is e.s.
- For tracking desired output $x_d$, the control law is:

$$v = x_d^{(n)} - k_0 e - k_1 \dot{e} - \ldots - k_{n-1} e^{(n-1)}$$

- $\therefore$ Exponentially convergent tracking, $e = x - x_d \to 0$.
- This method is extendable when the scalar $x$ was replaced by a vector and the scalar $b$ by an invertible square matrix.
- When $u$ is replaced by an invertible function $g(u) \rightsquigarrow u = g^{-1}(\frac{1}{b}[v - f])$,

# Feedback Linearization of a Two-link Robot

- A two-link robot: each joint equipped with
  - a motor for providing input torque
  - an encoder for measuring joint position
  - a tachometer for measuring joint velocity
- objective: the joint positions $q_l$ and $q_2$ follow desired position histories $q_{dl}(t)$ and $q_{d2}(t)$
- For example when a robot manipulator is required to move along a specified path, e.g., to draw circles.



A two-link robot

- Using the Lagrangian equations the robotic dynamics are:

$$\begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} + \begin{bmatrix} -h\dot{q}_2 & -h\dot{q}_2 - h\dot{q}_1 \\ h\dot{q}_1 & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} + \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

  where $q = [q_1 \;\; q_2]^T$: the two joint angles, $\tau = [\tau_1 \;\; \tau_2]^T$: the joint inputs, and

$$\begin{aligned} H_{11} &= m_1 l_c{}^2{}_1 + I_1 + m_2[l_1{}^2 + l_c{}^2{}_2 + 2\, l_1 l_{c2} \cos q_2] + \; I_2 \\ H_{22} &= m_2 l_c{}^2{}_2 + I_{2,} H_{12} = H_{21} = m_2 l_1 l_{c_2} \cos q_2 + m_2 l_c{}^2{}_2 + I_2 \\ g_1 &= m_1 l_{c1} \cos q_1 + m_2 g[l_{c2} \cos(q_1 + q_2) + l_1 \cos q_1] \\ g_2 &= m_2 l_{c2} g \cos(q_1 + q_2), \;\; h = m_2 l_1 l_{c2} \sin q_2 \end{aligned}$$

- Control law for tracking, (computed torque):

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} -h\dot{q}_2 & -h\dot{q}_2 - h\dot{q}_1 \\ h\dot{q}_1 & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix} + \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$

  where $v = \ddot{q}_d - 2\lambda\dot{\tilde{q}} - \lambda^2\tilde{q}$, $\tilde{q} = q - q_d$: position tracking error, $\lambda$: pos. const.
- $\therefore \; \ddot{\tilde{q}} \; + 2\lambda\dot{\tilde{q}} + \lambda^2\tilde{q} = 0$ where $\tilde{q}$ converge to zero exponentially.
- This method is applicable for arbitrary # of links

# Input-State Linearization

- When the nonlinear dynamics is **not** in a controllability canonical form, use algebraic transformations
- Consider the SISO system

$$\dot{x} = f(x, u)$$

- In **input-state linearization** technique:
  1. finds a state transformation $z = z(x)$ and an input transformation $u = u(x, v)$ s.t. the nonlinear system dynamics is transformed into $\dot{z} = Az + bv$
  2. Use standard linear techniques (such as pole placement) to design $v$.

# Example

- Consider
$$\begin{aligned}\dot{x}_1 &= -2x_1 + ax_2 + \sin x_1 \\ \dot{x}_2 &= -x_2 \cos x_1 + u \cos(2x_1)\end{aligned}$$
  - Equ. pt. $(0,0)$
  - The nonlinearity cannot be directly canceled by the control input $u$
  - Define a new set of variables:
$$\begin{aligned} z_1 &= x_1 \\ z_2 &= ax_2 + \sin x_1 \\ \therefore \; \dot{z}_1 &= -2z_1 + z_2 \\ \dot{z}_2 &= -2z_1 \cos z_1 + \cos z_1 \sin z_1 + au \cos(2z_1)\end{aligned}$$
  - The Equ. pt. is still $(0,0)$.
  - The control law: $u = \frac{1}{a\cos(2z_1)}(v - \cos z_1 \sin z_1 + 2z_1 \cos z_1)$
  - The new dynamics is linear and controllable: $\dot{z}_1 = -2z_1 + z_2, \quad \dot{z}_2 = v$
  - By proper choice of feedback gains $k_1$ and $k_2$ in $v = -k_1 z_1 - k_2 z_2$, place the poles properly.
  - Both $z_1$ and $z_2$ converge to zero, $\rightsquigarrow$ the original state $x$ converges to zero

- The result is **not global**.
  - The result is not valid when $x_l = (\pi/4 \pm k\pi/2),\ k = 0, 1, 2, ...$
- The input-state linearization is achieved by a **combination** of a state transformation and an input transformation with state feedback used in both.
- To **implement** the control law, the new states $(z_1, z_2)$ must be available.
  - If they are not physically meaningful or measurable, they should be computed by measurable original state $x$.
- If there is uncertainty in the model, e.g., on $a \rightsquigarrow$ error in the computation of new state $z$ as well as control input $u$.
- For **tracking control**, the desired motion needs to be expressed in terms of the new state vector.
- **Two questions** arise for more generalizations which will be answered in next lectures:
  - What classes of nonlinear systems can be transformed into linear systems?
  - How to find the proper transformations for those which can?

# Input-Output Linearization

- Consider

$$\begin{aligned} \dot{x} &= f(x, u) \\ y &= h(x) \end{aligned}$$

- Objective: tracking a desired trajectory $y_d(t)$, while keeping the whole state bounded
- $y_d(t)$ and its time derivatives up to a sufficiently high order are known and bounded.
- The difficulty: output $y$ is only *indirectly* related to the input $u$
  - $\therefore$ it is not easy to see how the input $u$ can be designed to control the tracking behavior of the output $y$.
- Input-output linearization approach:
  1. Generating a linear input-output relation
  2. Formulating a controller based on linear control

# Example

- Consider
$$\begin{aligned}\dot{x}_1 &= \sin x_2 + (x_2+1)x_3\\ \dot{x}_2 &= x_1^5 + x_3\\ \dot{x}_3 &= x_1^2 + u\\ y &= x_1\end{aligned}$$
- To generate a direct relationship between the output $y$ and the input $u$, differentiate the output $\dot{y} = \dot{x}_1 = \sin x_2 + (x_2+1)x_3$
- No direct relationship $\rightsquigarrow$ differentiate again: $\ddot{y} = (x_2+1)u + f(x)$, where $f(x) = (x_1^5 + x_3)(x_3 + \cos x_2) + (x_2+1)x_1^2$
- Control input law: $u = \frac{1}{x_2+1}(v - f)$.
- Choose $v = \ddot{y}_d - k_1 e - k_2 \dot{e}$, where $e = y - y_d$ is tracking error, $k_1$ and $k_2$ are pos. const.
- The closed-loop system: $\ddot{e} + k_2\dot{e} + k_1 e = 0$
- $\therefore$ e.s. of tracking error

# Example Cont'd

- The control law is defined everywhere except at **singularity points** s.t. $x_2 = -1$
- To **implement** the control law, full state measurement is necessary, because the computations of both the derivative $y$ and the input transformation need the value of $x$.
- If the output of a system should be differentiated $r$ times to generate an explicit relation between $y$ and $u$, the system is said to have relative degree $r$.
  - For linear systems this terminology expressed as $\#$ poles $-\#$ zeros.
- For any **controllable** system of order $n$, by taking at most $n$ differentiations, the control input will appear to any output, i.e., $r \leq n$.
  - If the control input never appears after more than $n$ differentiations, the system would not be controllable.

# Feedback Linearization

- **Internal dynamics**: a part of dynamics which is unobservable in the input-output linearization.
  - In the example it **can be** $\dot{x}_3 = x_1^2 + \frac{1}{x_2+1}(\ddot{y}_d(t) - k_1 e - k_2 \dot{e} - f)$
- The desired performance of the control based on the reduced-order model depends on the stability of the internal dynamics.
  - stability in BIBO sense
- **Example:** Consider

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2^3 + u \\ u \end{bmatrix} \qquad (2)$$
$$y = x_1$$

- Control objective: $y$ tracks $y_d$.
  - First differentiations of $y \rightsquigarrow$ linear I/O relation
  - The control law $u = -x_2^3 - e(t) + \dot{y}_d(t) \rightsquigarrow$ exp. convergence of $e$ : $\dot{e} + e = 0$
  - Internal dynamics: $\dot{x}_2 + x_2^3 = \dot{y}_d - e$
  - Since $e$ and $\dot{y}_d$ are bounded ($\dot{y}_d(t) - e \leq D$) $x_2$ is ultimately bounded.

- I/O linearization can also be applied to **stabilization** ($y_d(t) \equiv 0$):
  - For previous example the objective will be $y$ and $\dot{y}$ will be driven to zero and stable internal dynamics guarantee stability of the whole system.
  - No restriction to choose physically meaningful $h(x)$ i n $y = h(x)$
  - Different choices of output function leads to different internal dynamics which some of them may be unstable.
- When the **relative degree** of a system is the same as its **order**:
  - There is **no internal dynamics**
  - The problem will be **input-state linearization**

# Summary

- **Feedback linearization** cancels the nonlinearities in a nonlinear system s.t. the closed-loop dynamics is in a linear form.
- **Canceling the nonlinearities** and imposing a desired linear dynamics, can be applied to a class of nonlinear systems, named companion form, or controllability canonical form.
- When the nonlinear dynamics is **not** in a controllability canonical form, input-state linearization technique is employed:
  1. Transform input and state into companion canonical form
  2. Use standard linear techniques to design controller
- For **tracking** a desired traj, when $y$ is not directly related to $u$, I/O linearizaton is applied:
  1. Generating a linear input-output relation (take derivative of $y$ $r \leq n$ times)
  2. Formulating a controller based on linear control
- Relative degree: $\#$ of differentiating $y$ to find explicate relation to $u$.
- If $r \neq n$, there are $n - r$ internal dynamics that their stability be checked.

# Internal Dynamics of Linear Systems

- In general, directly determining the stability of the internal dynamics is not easy since it is nonlinear, nonautonomous, and coupled to the "external" closed-loop dynamics.
- We are seeking to translate the concept of internal dynamics to the more familiar context of linear systems.
- **Example:** Consider the controllable, observable system

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} x_2 + u \\ u \end{bmatrix} \\ y &= x_1 \end{aligned} \tag{3}$$

- Control objective: $y$ tracks $y_d$.
  - First differentiations of $y \rightsquigarrow \dot{y} = x_2 + u$
  - The control law $u = -x_2 - e(t) + \dot{y}_d(t) \rightsquigarrow$ exp. convergence of $e : \dot{e} + e = 0$
  - Internal dynamics: $\dot{x}_2 + x_2 = \dot{y}_d - e$
  - $e$ and $\dot{y}_d$ are bounded $\rightsquigarrow$ $x_2$ and therefore $u$ are bounded.

- Now consider a little different dynamics

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 + u \\ -u \end{bmatrix} \tag{4}$$
$$y = x_1$$

- using the same control law yields the following internal dynamics

$$\dot{x}_2 - x_2 = e(t) - \dot{y}_d$$

- Although $y_d$ and $y$ are bounded, $x_2$ and $u$ diverge to $\infty$ as $t \to \infty$
- why the same tracking design method yields different results?
  - Transfer function of (3) is: $W_1(s) = \frac{s+1}{s^2}$.
  - Transfer function of (4) is: $W_2(s) = \frac{s-1}{s^2}$.
  - $\therefore$ Both have the same poles but different zeros
  - The system $W_1$ which is minimum-phase tracks the desired trajectory perfectly.
  - The system $W_2$ which is nonminimum-phase requires infinite effort for tracking.

# Internal Dynamics

- Consider a third-order linear system with one zero

$$\dot{x} = Ax + bu, \quad y = c^T x \tag{5}$$

- Its transfer function is: $y = \frac{b_0 + b_1 s}{a_0 + a_1 s + a_2 s^2 + s^3} u$
- First transform it into the companion form:

$$\begin{aligned} \begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \\ y &= [b_0 \ b_1 \ 0] \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \end{aligned} \tag{6}$$

- In second derivation of $y$, $u$ appears:
  $\ddot{y} = b_0 z_3 + b_1(-a_0 z_1 - a_1 z_2 - a_2 z_3 + u)$
- $\therefore$ Required number of differentiations (the relative degree) is indeed the same as # of poles- # of zeros
  - Note that: the I/O relation is independent of the choice of state variables $\rightsquigarrow$ two differentiations is required for u to appear if we use (5).
- The control law: $u = (a_0 z_1 + a_1 z_2 + a_2 z_3 - \frac{b_0}{b_1} z_3) + \frac{1}{b_1}(-k_1 e - k_2 \dot{e} + \ddot{y}_d)$
- $\therefore$ an exp. stable tracking is guaranteed
- The internal dynamics can be described by only one state equation
  - $z_1$ can complete the state vector,( $z_1$, $y$, and $\dot{y}$ are related to $z_1$, $z_2$ and $z_3$ through a one-to-one transformation).
  - $\dot{z}_1 = z_2 = \frac{1}{b_1}(y - b_0 z_1)$
  - $y$ is bounded $\rightsquigarrow$ stability of the internal dynamics depends on $-\frac{b_0}{b_1}$
  - If the system is minimum phase the internal dynamics is stable (independent of initial conditions and magnitude of desired trajectory)

# Zero-Dynamics

- For linear systems the stability of the internal dynamics is determined by the locations of the zeros.
- To extend the results for nonlinear systems the concept of zero should be modified.
- Extending the notion of zeros to nonlinear systems is not trivial
  - In linear systems I/O relation is described by transfer function which zeros and poles are its fundamental components. **But** in nonlinear systems we cannot define transfer function
  - Zeros are intrinsic properties of a linear plant. **But** for nonlinear systems the stability of the internal dynamics may depend on the specific control input.
- Zero dynamics: is defined to be the internal dynamics of the system when the system output (and all of its derivatives up to r-1) is **kept** at zero by the input.

- For dynamics (2), the zero dynamics is $\dot{x}_2 + x_2^3 = 0$
  - we find input $u$ to maintain the system output at *zero uniquely* (keep $x_1$ zero in this example),
  - By Lyap. Fcn $V = x_2^2$ it can be shown it is a.s
- For linear system (5), the zero dynamics is $\dot{z}_1 + (b_0/b_1)z_1 = 0$
- $\therefore$ The poles of the zero-dynamics are exactly the zeros of the system.
- In **linear systems**, if all zeros are in LHP $\rightsquigarrow$ g.a.s. of the zero-dynamics $\rightsquigarrow$ g.s. of internal dynamics.
- In **nonlinear systems**, no results on the global stability
  - only local stability is guaranteed for the internal dynamics even if the zero-dynamics is g.e.s.
- Zero-dynamics is an intrinsic feature of a nonlinear system, which does not depend on the choice of control law or the desired trajectories.
- Examining the **stability** of zero-dynamics is easier than examining the stability of internal dynamics, **But** the result is local.
  - Zero-dynamics only involves the internal states
  - Internal dynamics is coupled to the external dynamics and desired trajs.

# Zero-Dynamics

- Similar to the linear case, a nonlinear system whose zero dynamics is asymptotically stable is called an asymptotically minimum phase system,
- If the **zero-dynamics is unstable**, different control strategies should be sought
- As summary control design based on input-output linearization is in three steps:
  1. Differentiate the output $y$ until the input $u$ appears
  2. Choose $u$ to cancel the nonlinearities and guarantee tracking convergence
  3. Study the stability of the internal dynamics
- If the relative degree associated with the input-output linearization is the same as the order of the system $\rightsquigarrow$ the nonlinear system is fully linearized $\rightsquigarrow$ satisfactory controller
- Otherwise, the nonlinear system is only partly linearized $\rightsquigarrow$ whether or not the controller is applicable depends on the stability of the internal dynamics.